

ML – Regression – Boosting Algorithms – Assignment

AdaBoost (Adaptive Boosting)

- AdaBoost is a machine learning boosting algorithm used to improve prediction accuracy.
- It combines multiple weak models (usually decision trees) into one strong model.
- It gives more importance to the data points that were predicted wrongly.
- Each new model learns from the previous model's mistakes.
- Final prediction is made by combining all model outputs with weighted voting.

XGBoost (Extreme Gradient Boosting)

- XGBoost is an advanced and high-performance boosting algorithm.
- It builds decision trees sequentially to correct previous errors.
- It uses gradient boosting techniques for faster and more accurate predictions.
- XGBoost is popular because it is fast, scalable, and handles overfitting well.
- It is widely used in machine learning competitions and real-world AI projects.

LightGBM (Light Gradient Boosting Machine)

- LightGBM is a fast-boosting algorithm developed by Microsoft.
- It is designed for high speed and low memory usage.
- Unlike traditional boosting, it grows trees leaf-wise for better accuracy.
- LightGBM works very well with large datasets and complex data.
- It is commonly used in AI, data science, and big data applications because of its speed.

Simple Interview Line

- AdaBoost → Focuses more on wrong predictions.
- XGBoost → Faster and more optimized boosting algorithm.
- LightGBM → Very fast and memory-efficient boosting model for large datasets.

